CS 330, Fall 2026, Homework 9
Due Wednesday, November 4, 2026

Guidelines. *The same homework guidelines apply as for Homework 1.*

1. Recurrences

Consider the following algorithm A:

Algorithm 1: A(n)

Input: a non-negative integer n

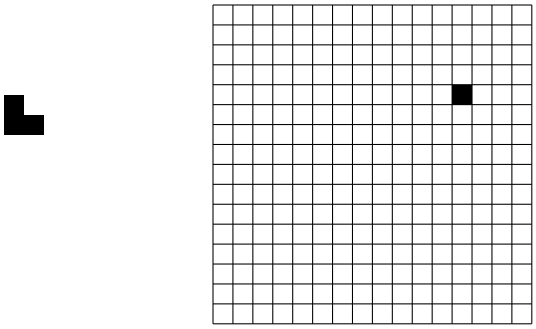
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1 if n = 0 then
2     return 0
3 if n is even then
4     return A(n/2)
5 else
6     return A(2(n − 1)) + 1
```

(a) Prove that A terminates when its input n is a non-negative integer
(b) Write a recurrence for the running time (in terms of n) of A.
(c) Give a closed form for its asymptotic running time (using whichever method you like).
(d) If we look at the tree of recursive calls to A made by this algorithm, what is its depth (asymptotically, in terms of n)? How many leaves does it have (asymptotically, in terms of n)?
(e) What function of n does A compute? It's easy to express it in terms of the binary expansion of n (i.e. the bits you get when you write n in base 2).

2. Carpets

You finally graduated from BU and took a job that allowed you to afford your very own apartment. Congratulations! The living space in your new apartment is a $2^k \times 2^k$-foot square (which you mentally divide into 2^{2k} cells, each 1 foot square). Bizarrely, exactly one cell of the floor is covered by carpet, and the rest is bare cement (Figure 1 (b)). Fortunately, a friend has given you a pile of $\frac{2^{2k}-1}{3}$ carpet pieces. Each carpet piece is L shaped, formed by three 1-by-1 adjacent squares (Figure 1(a)). Each piece covers 3 cells, so you have just enough carpet to cover the whole cement part of the floor, if the pieces can fit. Your job is to find a way to lay the pieces so they cover the cement. The carpet should cover all cells except the covered one with no overlaps. You may not cut the carpet pieces—you have to use them as they are—but it is fine to rotate them.

(a) Design a divide-and-conquer algorithm for this problem. The inputs are $n = 2^k$ (which determines the size of the room) and the coordinates $(x, y) \in \{1, ..., n\} \times \{1, ..., n\}$ of the covered square. You may assume n is an integer power of 2. The output should be a list of triples, where each triple describes the position of one of the carpet pieces you will put down. An example of a correct covering of the square is illustrated in figure 2. The caption shows what your algorithm should output given this solution.



(a) (b)

Figure 1: (a) A carpet piece in one of its 4 possible orientations. (b) A 16×16 floor with one covered cell.

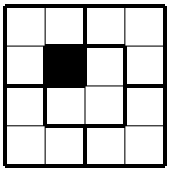


Figure 2: A 4×4 floor with covered cell $(x, y) = (2, 2)$, tiled by five carpet pieces. Taking $(1, 1)$ as the top-left cell, with x increasing to the right and y increasing downward, the algorithm outputs the list $[(3, 3, \text{TLdirection}),\ (1, 4, \text{TRdirection}),\ (1, 1, \text{BRdirection}),\ (4, 4, \text{TLdirection}),\ (4, 1, \text{BLdirection})]$. Each triple gives the cell at the bend of an L-shaped piece, followed by its orientation: XYdirection means the piece leaves the XY cell of its 2×2 bounding square empty.

First, explain your algorithm concisely in English (feel free to use pictures). Second, specify your algorithm using clear pseudocode.

(b) Prove that your algorithm is correct.

(c) Give a recurrence for the worst case running time of your algorithm in terms of n and solve it. How long does your algorithm take as a function of n?